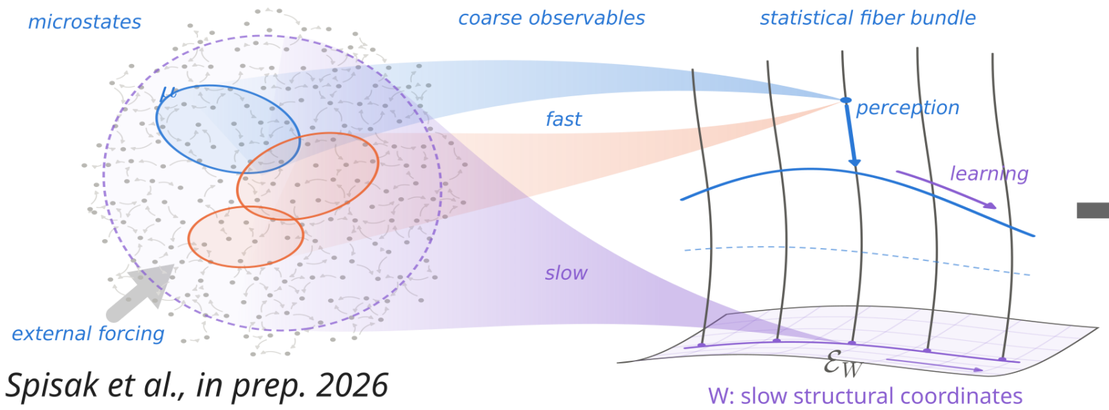
Line A "Approach and Theory": first principles & falsifiable predictions

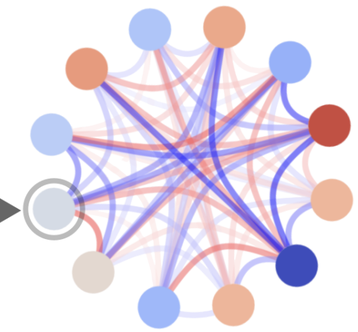
A physical account for perception and learning



W: slow structural coordinates

Spisak et al., in prep. 2026

Entropic (self-orthogonalizing) attractor networks



Spisak & Friston, 2026

Learning

$$\Delta J_{ij} \propto \underbrace{\sigma_i \sigma_j}_{\text{observed correlation (Hebbian)}} - \underbrace{L\left(b_i + \sum_{k \neq i} J_{ik}\,\sigma_k\right)\sigma_j}_{\text{predicted correlation (anti-Hebbian)}}$$

- predictive coding
- self-orthogonalization
- separates latent causes

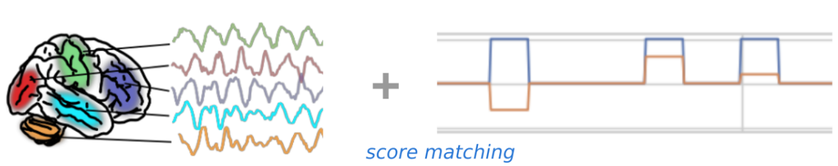
Inference

$$\mathbb{E}_q[\sigma_i] = L(b_q) = L\Bigg(\underbrace{\underbrace{b_i}_{\text{bias}} + \underbrace{\sum_{j \neq i} J_{ij}\sigma_j}_{\text{weighted input}}}_{\text{sigmoid (Langevin)}}\Bigg)$$

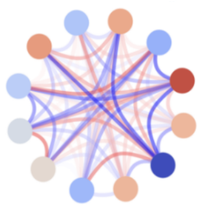
- Bayesian belief update
- attractor states: priors
- MCMC sampling from the posterior
- continual and sequence learning

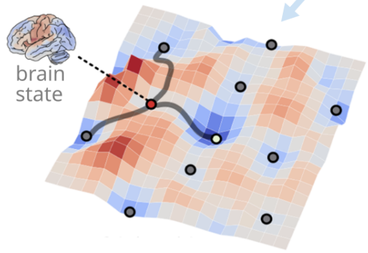
Line B "Brains": Computational generative brain modelling

Task and resting fMRI dynamics

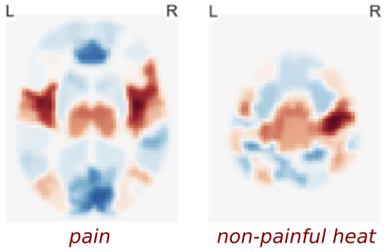


score matching





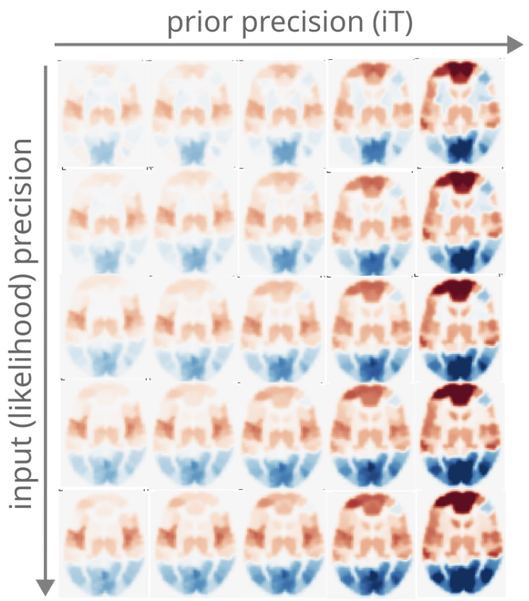
Weights:
Network level dynamics



pain *non-painful heat*

Biases:
Upstream mesoscale dynamics

Task-related activations as Bayesian inference

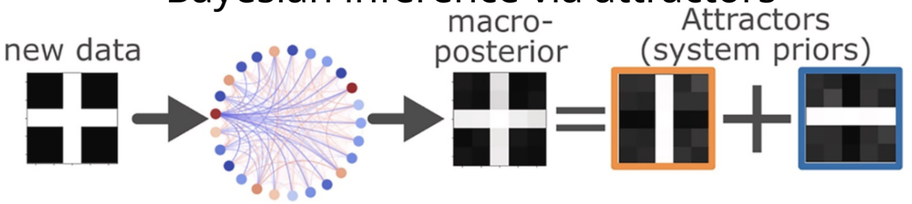


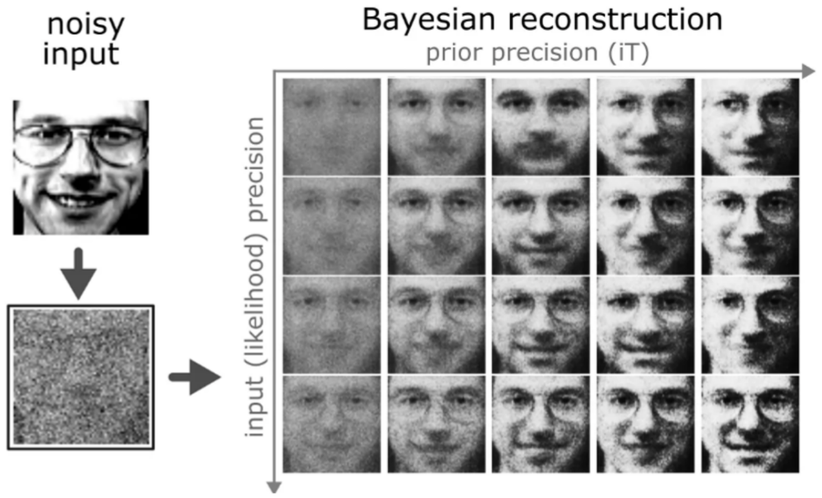
posterior activation

Englert et al., 2026; python package attractome

Line C "Computers": Emergent learning as an alternative route to AI

Bayesian inference via attractors





Spisak and Friston, 2026; Spisak et al. in prep. 2026